Akoizo Original Practice - 2027 Rotation Form 2

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Hydrophobic attraction is a new covalent bond?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Always

B. Only between leucines

C. Only in water

D. No

Answer: _____

2. (2 points) HA contributes to…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Attachment and fusion

B. Only genome replication

C. Only salt sensing

D. Only ATP storage

Answer: _____

3. (2 points) A strong annotation explains…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Only material cost

B. Identity, role, and evidence

C. Only a color

D. Only a teacher name

Answer: _____

4. (2 points) A disulfide links…

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Any two glycines

B. DNA bases

C. Two phosphate groups

D. Cysteines

Answer: _____

5. (2 points) Reassortment exchanges…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Genome segments

B. Only individual atoms

C. Only amino-acid colors

D. Only antibodies

Answer: _____

6. (2 points) A missing segment should be…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Ignored in numbering

B. Explicitly marked or qualified

C. Silently invented

D. Called observed

Answer: _____

7. (2 points) More geometric contacts prove greater stability?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. Always

B. Only in cartoons

C. Only for enzymes

D. No

Answer: _____

8. (2 points) The 2009 outbreak is used here as…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. A historical case study

B. A current case count

C. A permanent event assignment

D. A diagnosis guide

Answer: _____

9. (2 points) Model review should use…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. An unrelated year automatically

B. The verified applicable rubric

C. Invented point values

D. Only appearance

Answer: _____

10. (4 points) Why does pH matter?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

Answer: _____
_____
_____

11. (2 points) The assigned interval runs from residue 153 through 252, inclusive. How many numbered residue positions are included?

Answer: _____
_____
_____

12. (2 points) At the specified scale of 2 cm per residue, what backbone material length corresponds to 100 residues before folding?

Answer: _____
_____
_____

13. (2 points) State the assigned 2027 structure identifier, chain and residue interval, and distinguish these identifiers from a claim that all residues are experimentally resolved.

Answer: _____
_____
_____

14. (2 points) Explain the functional relevance of hemagglutinin binding to sialic-acid receptors in influenza entry.

Answer: _____
_____
_____

15. (2 points) Why does identifying an alpha helix require backbone geometry rather than simply finding a hydrophobic sequence stretch?

Answer: _____
_____
_____

16. (2 points) A synthetic structure worksheet places two oppositely charged side chains close together. Describe a possible stabilizing interaction and a qualification.

Answer: _____
_____
_____

17. (2 points) Explain why displaying a ligand-binding pocket without identifying chain and residue numbering can mislead a model reviewer.

Answer: _____
_____
_____

18. (2 points) What is the distinction between a displayed molecular contact and a proven functional interaction?

Answer: _____
_____
_____

19. (2 points) A model includes a decorative feature unrelated to protein structure or function. Explain why it does not improve its scientific explanation.

Answer: _____
_____
_____

20. (2 points) Explain why rotating a JUDE molecular view does not alter the primary sequence or chemical connectivity.

Answer: _____
_____
_____